

Employee Leave Management System

Business Requirements & Technical Solution Design

Prepared for: NovaTech Manufacturing GmbH

SAP CAP / SAP Fiori Elements / SAP BTP Cloud Foundry

Document version 1.0

Prepared as project reference for CAP implementation

1. Business Context

NovaTech Manufacturing GmbH is a mid-sized industrial equipment manufacturer (~1,200 employees) running SAP S/4HANA on-premise for core Finance and HR, and adopting SAP BTP for side-by-side extension applications.

Employees currently request leave via email or paper forms. HR tracks balances manually in a shared spreadsheet, with no audit trail and no visibility into team availability. The goal is a modern, BTP-hosted application where employees submit leave requests, managers approve them, and HR maintains full oversight — accessible from any device with a consistent SAP Fiori look and feel.

Stakeholders

Role	Description	Primary needs
Employee	Any staff member	Submit requests, view own history/balance, cancel pending requests
Manager	Line manager of a team	View/approve team requests, see team calendar
HR Admin	HR department staff	Full visibility, manage leave types, override decisions, reporting

2. Functional Requirements

ID	Requirement	Description
FR-01	Leave request submission	Employee creates a request with leave type, start date, end date, optional reason. System auto-calculates number of days requested.
FR-02	Leave balance tracking	Each employee has a balance per leave type. Submission is blocked if requested days exceed remaining balance.
FR-03	Approval workflow	Submitted requests go to the employee's manager with status Pending. Manager can Approve or Reject with an optional comment. Approval deducts balance; rejection does not.
FR-04	Request cancellation	Employee can cancel a Pending request. Cancelling an Approved request requires HR Admin intervention.
FR-05	Leave types master data	HR Admin maintains configurable leave types (Annual, Sick, Unpaid, etc.), each with attributes such as requiresApproval.
FR-06	Team visibility (Manager)	Manager sees a list/calendar of direct reports' approved and pending leave.
FR-07	HR oversight	HR Admin can view, filter, and edit any leave request company-wide, and manage employee leave balances.
FR-08	Status tracking and audit	Every request has a status (Draft, Pending, Approved, Rejected, Cancelled). System retains who approved/rejected and when.

3. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	SAP Fiori Elements UI, consistent with SAP Fiori design guidelines, usable on tablet/mobile.
NFR-02	Security	Role-based authorization enforced at the service layer, not only in the UI.
NFR-03	Auditability	Standard CAP audit fields (createdBy, createdAt, modifiedBy, modifiedAt) on all business entities.
NFR-04	Maintainability	Clean separation of data model, service layer, and business logic per CAP best practice.
NFR-05	Deployability	Runs on SAP BTP, Cloud Foundry, packaged as a Multi-Target Application (MTA).

4. Scope

In scope

- Leave request CRUD, approval workflow, balance tracking
- Three-role authorization: Employee, Manager, HR Admin
- SAP Fiori Elements UI (List Report – Object Page floorplan)
- Deployment to SAP BTP Cloud Foundry

Out of scope for this build

- Live integration with SAP HCM / Employee Central (employees are local master data)
- Email / notification integration
- Public holiday calendar integration
- Multi-country / multi-language leave policy variants

Assumptions

- Single-tenant application, not a multi-tenant SaaS build
- Users are assigned roles via BTP role collections, not sourced live from an HR system
- One leave year = calendar year; balances reset annually (model supports it, automation not built)

5. Solution Architecture

Browser (Fiori Launchpad) → App Router (routing + session auth) → CAP Service (Node.js, OData v4 + business logic) → Database. XSUAA issues OAuth2/JWT tokens carrying the user's role scopes, checked by the App Router and the CAP service.

Key technology decisions

Decision	Choice	Rationale
CAP runtime	Node.js (@sap/cds)	Faster local dev cycle, larger community, npm ecosystem
Local database	SQLite	Zero setup, instant iteration during development
Production database	SAP HANA Cloud	Standard enterprise target for productive BTP CAP apps
UI technology	SAP Fiori Elements (List Report – Object Page)	Generated from CDS annotations, SAP standard for master-detail + workflow
Authentication	XSUAA	Standard BTP identity service, integrates with CDS @requires/@restrict
Deployment packaging	MTA (Multi-Target Application)	Standard packaging for multi-module BTP apps

6. Conceptual Data Model

Entities and key fields

Entity	Key fields	Notes
Employee	ID, name, email, managerID (self-reference)	managerID references another Employee — models the org hierarchy without a separate Manager entity
LeaveType	ID, name, requiresApproval	Master data maintained by HR Admin, not hardcoded
LeaveRequest	ID, employeeID, leaveTypeID, startDate, endDate, days, status	status: Draft, Pending, Approved, Rejected, Cancelled
LeaveBalance	ID, employeeID, leaveTypeID, totalDays, usedDays	Separate from LeaveRequest — a stateful running total, adjustable by HR Admin directly

Relationships

- Employee 1—many LeaveRequest (submits)
- Employee 1—many LeaveBalance (has)
- LeaveType 1—many LeaveRequest (classifies)
- LeaveType 1—many LeaveBalance (tracks)

- Employee 1—many Employee (manages, self-referencing via managerID)

7. Naming & Project Conventions

Item	Convention	Value
Project folder	kebab-case	leave-management
CDS namespace	reverse-domain	com.novatech.leavemgmt
Entity names	PascalCase, singular	Employee, LeaveRequest, LeaveType, LeaveBalance
Service name	PascalCase + Service	LeaveService
Field names	camelCase	startDate, employeeID, requiresApproval
Fiori app module	kebab-case	leave-request-manage

Standard CAP folder structure

leave-management/ → app/ (UI apps), db/ (schema.cds + sample data), srv/ (leave-service.cds + leave-service.js), package.json, mta.yaml (deployment), xs-security.json (XSUAA roles/scopes).

Authorization concept

BTP role collection	Scope name	Access
LeaveEmployee	Employee	CRUD on own leave requests only
LeaveManager	Manager	Read/approve direct reports' requests
LeaveHRAdmin	HRAdmin	Full read/write across all data, manage master data

This document consolidates the requirements and design decisions established during solution design. It is intended as the reference brief for implementation.